Malaysia Sovereign AI & Digital Infrastructure Grid

Master Reference Registry & Regional Node Blueprint (2026)

1. Sovereign Registry Filings (Act 834)

Official geographical indication registrations filed under the Intellectual Property Corporation of Malaysia (MyIPO) for sovereign computing clusters.

Parameter	Node Description 1: Sazwan Digital Valley	Node Description 2: Kedah AI Nexus
Filing Code	GI-2026-07734 (Pending)	GI-2026-07735 (Pending)
Catalog ID	KDH-SDV-2026-001	KDH-KAN-2026-002
Physical Focus	Sungai Petani, Kedah (AI R&D Hub)	Northern Corridor, MY Cloud Infra
Core Engine	SGAE 2.0 Engine Development & Decentralized Identity (DID)	Sovereign Cloud Nodes & Key Management (KMS)
Filing Date	08/04/2026	12/04/2026
SHA-256 Digest	32D8A53...	32D8A53...

2. Strategic National Corridors & Hubs

Cross-border backbone connectivity map aligning regional R&D nodes with Tier-1 infrastructure networks:

Region / State	Infrastructural Role & Core Capability	Key Facilities & Frameworks
Kuala Lumpur	Central Command, Tier-1 Global Node backbone, Oracle Sovereign Cloud Center ($6.5B investment).	MDEC HQ (Bangsar South), WORQ & Common Ground MD Hubs
Selangor	Hyperscale Computing Cluster, Tier IV data handling, autonomous systems testing.	Cyberjaya Hubs, NEXTDC KL1 (65MW Tier IV Facility)
Kedah (Northern)	Primary Sovereign AI R&D Engine Development, semiconductor design integration.	Sungai Petani AI Cluster, Kulim Hi-Tech Park (KHTP)
Johor (Southern)	Green Mega-Cloud Ecosystem, international gateway linking Singapore cross-border routing.	Sedenak Tech Park (STPi), YTL Green Data Center Park

Sarawak & Sabah	East Malaysia Sovereignty, Green Hydrogen powered architectures, BIMP-EAGA gateways.	Kuching Tech Hubs, SMA Blueprints, Kota Kinabalu Gateways

3. Technical Architecture: SGAE 2.0 Integration Loop

The network workflow executes high-throughput cryptographic verification utilizing a split-node architecture. The Application Layer (Lembah Klang) processes concurrent user token validations, while the Core Cryptographic Engine (Sungai Petani Cluster) manages completely isolated data encryption, key rotation (KMS), and immutable state storage validated via localized SHA-256 digital notary digests before routing outward through Tier-1 egress paths.